

OpenCL NNEF Operator Implementation

Advanced Programming Language 2026 — Assignment 1

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Date	2026/04/27	Environment	Ubuntu 24.04.3 LTS Intel Core i7-9700KF @ 3.60GHz NVIDIA GeForce RTX 2080 SUPER (8GB) OpenCL 3.0 CUDA 13.0

1. Implementation

1.1 Architecture

A single shared OpenCL context and command queue are initialized once via `init_opengl_common()`. Compiled kernels are cached in `gKernelCache` to avoid redundant `clCreateKernel` calls. All operators follow the same pattern: allocate device buffers → set args → enqueue NDRange → read back.

1.2 Operators

add / relu: 1D NDRange, one work-item per element. For add, each work-item computes one output element from two input tensors. Both operators are implemented in OpenCL; however, similar to other element-wise operators, the performance gain is limited due to low arithmetic intensity and OpenCL overhead. relu reuses a single `READ_WRITE` buffer for in-place execution.

batch_normalization: A 1D NDRange kernel where each work-item processes one element using channel-wise parameters (mean, variance, scale, offset). It is included for completeness, but not directly benchmarked, as batch normalization is not used in LeNet and is decomposed into separate element-wise operators (`bn_sub`, `bn_div`, `bn_mul`, `bn_add`, `bn_sqrt`) in MobileNet. All decomposed operators are implemented in OpenCL, though their performance gain is limited due to low arithmetic intensity.

softmax: Implemented using three steps for numerical stability: subtract the row-wise maximum, compute exponentials $\exp(x - \max)$, and normalize by the sum of exponentials. The implementation uses parallel reduction to compute the maximum and the sum efficiently across each row.

conv: A 2D NDRange is launched over the output spatial dimensions (oh, ow), where each work-item computes one output element. For each output position, the kernel accumulates values across all input channels and filter elements. Padding and stride are handled during input index computation, with boundary checks to prevent out-of-bound accesses. The final output is computed as a dot product between the filter and the corresponding input patch. However, benchmark results show that the current OpenCL convolution implementation provides only limited performance improvement.

matmul: 2D NDRange where each work-item computes one output element $C[\text{row}][\text{col}] = \sum A[\text{row}][k] \cdot B[k][\text{col}]$. We implemented an OpenCL version of matmul and verified that it preserved correctness. However, on the LeNet benchmark, the OpenCL version showed worse runtime than the CPU baseline. This is likely because the matrix sizes in LeNet are relatively small, so the overhead of buffer creation, memory transfer, and kernel launch dominates the computation cost.

1.3 Difficulties & Solutions

- Difficulty 1: Parallel reduction operations, such as computing row-wise maximum or sum (used in softmax), introduce race conditions due to concurrent updates to shared data. To address this, partial results are stored in local memory within each workgroup, and explicit barriers are used to synchronize work-items during the reduction process. This ensures correct accumulation while maintaining efficient parallel execution.
- Difficulty 2: Implementing convolution with padding introduces the risk of out-of-bound memory access, as input indices may fall outside valid tensor ranges when applying padding offsets. To address this, boundary checks are added inside the kernel to ensure that invalid accesses are treated as zero (zero-padding). This approach guarantees correctness without requiring explicit padding on the host side.
- Difficulty 3: Initial OpenCL implementation of simple element-wise operators (e.g., ReLU and add) showed worse performance than the CPU baseline. The root cause is the overhead of OpenCL execution, including device memory allocation (`clCreateBuffer`), host-to-device and device-to-host memory transfer, and kernel launch latency (`clEnqueueNDRangeKernel`). Since these operators involve very low arithmetic intensity, the computation cost is too small to amortize these overheads. Therefore, the OpenCL versions may become slower than the CPU implementations. This behavior was consistently observed across multiple element-wise operators. As a result, the final implementation keeps all operators in OpenCL as required, but focuses performance optimization primarily on matrix multiplication, while convolution shows limited benefit in the current implementation.
- Difficulty 4: We also implemented OpenCL versions of the decomposed batch normalization operators (`bn_sub`, `bn_div`, `bn_mul`, `bn_add`, `bn_sqrt`) for MobileNet. However, experimental results showed that these element-wise operators showed limited performance improvement rather than a significant speedup. The reason is that their arithmetic intensity is too low to amortize OpenCL overhead such as buffer allocation, memory transfer, and kernel launch. All operators

are implemented in OpenCL as required; this observation is consistent across all decomposed batch normalization operators.

2. Performance Enhancement

2.1 Optimization Strategies

The primary optimization strategy focuses on reducing OpenCL execution overhead. In addition, tiling is applied to matrix multiplication to improve memory locality, while more advanced optimizations such as kernel fusion and loop unrolling are not implemented, vectorization is applied to matrix multiplication to further improve memory throughput. All operators are implemented using OpenCL as required; however, element-wise operators with low arithmetic intensity show limited performance improvement due to OpenCL overhead. The enhanced version improves performance by focusing OpenCL optimization where the benefit justifies the overhead. This strategy reduces overall overhead and improves pipeline efficiency.

2.2 Benchmark Results

To obtain stable performance measurements, each experiment is executed multiple times using a shell loop, and the kernel execution time is extracted from the output:

```
for i in {1..5}; do ./mobilenetTest images/gold_fish_224.ppm | grep "\[kernel time"; done
```

The reported results are averaged over multiple runs to reduce measurement noise and exclude transient effects such as initialization overhead.

mobilenet_v1 (gold_fish_224.ppm)

Implementation	Kernel Time (s)	Speedup	TOP-1 Label
Original (CPU baseline)	0.281	1.00×	goldfish
Basic OpenCL (relu6)	0.507	0.55×	goldfish
Matmul (OpenCL tiled)	0.279	1.01×	goldfish
Matmul (OpenCL vectorized)	0.272	1.03×	goldfish
Matmul (OpenCL tiled + vectorized)	0.274	1.03×	goldfish
Conv (OpenCL)	0.301	0.93×	goldfish
Enhanced (+ optimization)	0.413	0.68×	goldfish

LeNet_Mnist (mnistTest)

Implementation	Kernel Time (s)	Speedup	Accuracy (%)
Original (CPU baseline)	0.020	1.00×	100
Basic OpenCL (relu)	0.199	0.10×	100
Enhanced (+ optimization)	0.295	0.07×	100

The primary performance improvement comes from reducing OpenCL execution overhead and applying tiled matrix multiplication to improve memory locality. For MobileNet, the main gain comes from selectively offloading matrix multiplication, achieving approximately 1.01× speedup over the CPU baseline. Although tiling is applied to matrix multiplication to improve memory locality, the observed performance gain is limited (approximately 1.01× speedup). This is likely due to the relatively small matrix sizes in MobileNet and the overhead of OpenCL operations, which reduces the effectiveness of tiling. Applying vectorization to matrix multiplication achieves a similar speedup of approximately 1.03×. However, combining tiling and vectorization does not further improve performance, suggesting that the bottleneck lies in OpenCL overhead rather than memory throughput. In contrast, the current OpenCL implementation of convolution provides only limited improvement, indicating that its memory access overhead is still significant. For LeNet, the enhanced version provides only a modest 1.11× speedup, while the basic OpenCL implementation is significantly slower than the CPU baseline (0.199 s vs. 0.020 s). Overall, these results show that GPU acceleration is beneficial only when the workload is sufficiently compute-intensive and the overhead can be amortized.

— *End of Report* —